TEAM XERXES · IIIT BHUBANESWAR · SIH26262

AnuPalan research dossier

Every news report, rule, tool and diagram we used to design AnuPalan, in one place, with what each source says and how it shaped the project.

29 sources in 6 sections | 8 project diagrams | Compiled 27 Oct 2026 | For research purposes only

00 The problem in numbers

Figures taken from the articles in this dossier.

7
students died in the Satya Niketan PG collapse, 6 Sep 2026
The Better India

8 of 2,453
Delhi PGs had a structural stability certificate
The Tribune

353
teachers listed full-time at 2+ colleges; two at 11 each
The News Minute

342
AICTE institutions closed in three years
Careers360

277
engineering colleges ran without AICTE approval (2018)
Careers360

13,861
faculty records flagged as suspicious in Tamil Nadu
Careers360

01 Student safety: buildings, hostels and PGs

Why AnuPalan tracks structural and fire certificates, and lets students report unsafe buildings.

LIVELAW · 7 SEP 2026

Satya Niketan PG Collapse: Delhi High Court Orders High Level MCD Enquiry, Directs Doubling Of Rescue Efforts

A five-storey student PG in Satya Niketan, Delhi, collapsed on 6 September 2026. Hearing a PIL, the High Court ordered the MCD to check whether the building had valid permission and to report on PG buildings and violations within 10 days.

How we used it: The lead example on slide 2. It shaped the hostel/PG watchlist and the structural-certificate alerts.

https://www.livelaw.in/high-court/delhi-high-court/satya-niketan-pg-collapse-delhi-high-court-orders-high-level-mcd-enquiry-directs-doubling-of-rescue-efforts-548950

THE BETTER INDIA · 11 SEP 2026

After the Delhi PG Collapse, Experts Suggest 8 Fixes for Safer Student Housing

Seven students died when the roughly 50-year-old building, recently converted into student housing, collapsed. Experts propose livability ratings, sensor-based structural monitoring, enforced structural certification and better space standards.

How we used it: Source for the death toll, and for treating structural certification as something to enforce, not just file.

https://thebetterindia.com/governance/delhi-pg-collapse-student-housing-safety-solutions-12516180

THE TRIBUNE · 20 SEP 2026

Every student deserves quality and safe hostel facility

A survey after the collapse found 1,724 of 2,453 Delhi PG accommodations were built without approved plans, and only eight had structural stability certificates.

How we used it: The clearest number for why certificate validity must be tracked live (requirement E).

https://www.tribuneindia.com/news/in-depth/every-student-deserves-quality-and-safe-hostel-facility/

ANI · 30 SEP 2026

Fresh PIL in Delhi HC seeks student safety mechanism for PGs after Satya Niketan collapse

The PIL asks for student accommodation monitoring committees, a 24×7 helpline, an online grievance portal, periodic safety checks of private student housing, and a full regulatory framework.

How we used it: Directly matches AnuPalan's public complaint site, helpline and Sahayak AI bot.

https://aninews.in/news/national/general-news/fresh-pil-in-delhi-hc-seeks-student-safety-mechanism-for-pgs-after-satya-niketan-collapse20260930132301/

FREE PRESS JOURNAL · 30 SEP 2026

Delhi HC Seeks Response On Student Safety In PGs, Rented Flats And Private Hostels After Satya Niketan Building Collapse

The court sought responses from the Delhi government, MCD, Delhi Police, Delhi University, GGSIP University, UGC and the Ministry of Education, with a hearing set for 14 October 2026.

How we used it: Shows how many bodies share responsibility, which is why AnuPalan routes each case to a named owner.

https://www.freepressjournal.in/education/delhi-hc-seeks-response-on-student-safety-in-pgs-rented-flats-and-private-hostels-after-satya-niketan-building-collapse

CAREERS360 · 25 FEB 2026

Delhi University: Classroom ceiling collapses in Dyal Singh College during class, no injuries reported

The ceiling of a portable cabin classroom fell during a BA History class. No one was hurt; the college blamed the age and poor condition of its temporary cabins.

How we used it: Basis for the 'ceiling cracks' complaint used in the Sahayak bot demo.

https://news.careers360.com/delhi-university-classroom-ceiling-collapses-in-du-dyal-singh-college-during-class-no-injuries-reported/amp

CAREERS360 · 14 APR 2024

Eight students injured in hostel building fire in Kota

A fire in a five-storey boys' hostel injured eight students. A fire officer said the building lacked fire safety measures and a fire clearance certificate.

How we used it: Why a lapsed fire NOC raises an alert before something goes wrong.

https://news.careers360.com/eight-students-injured-in-hostel-building-fire-in-kota

02 Unapproved and fake technical institutions

Why AnuPalan keeps a checkable record of approval status and a risk watchlist.

ANI · 28 JUL 2026

Delhi HC fake Universities PIL: AICTE says it has robust mechanism to identify, penalise unapproved technical institutions

AICTE told the court it publishes lists of institutions by approval status, declares institutions without approval as unapproved, starts penal proceedings, and can seek closure or ask states to act.

How we used it: The process AnuPalan's watchlist and audit trail are designed to speed up.

https://aninews.in/news/national/general-news/delhi-hc-fake-universities-pil-aicte-says-it-has-robust-mechanism-to-identify-penalise-unapproved-technical-institutions20260728113752/

CAREERS360 · 2 AUG 2018

277 fake engineering colleges functioning without AICTE approval, Delhi leads with 66

277 engineering colleges were running without AICTE approval: Delhi 66, Telangana 35, West Bengal 27, Karnataka 23, Uttar Pradesh 22.

How we used it: Data for the bar chart on slide 2.

https://news.careers360.com/277-fake-engineering-colleges-functioning-without-aicte-approval-delhi-leads-66

03 Faculty fraud: one teacher, many colleges

Why AnuPalan's proof scanner checks for documents and staff reused across institutions.

THE NEWS MINUTE · 24 JUL 2024

Same prof in 11 colleges: NGO exposes alleged fraud in Anna University affiliation

Arappor Iyakkam found 353 teachers shown as full-time faculty at more than one college in 2023-24; two professors were listed at 11 colleges each. It alleges 224 of 480 affiliated colleges were involved.

How we used it: Example on slide 2; basis for the 'unique, not reused' check in the AI proof scanner.

https://www.thenewsminute.com/tamil-nadu/same-prof-in-11-colleges-ngo-exposes-alleged-fraud-in-anna-university-affiliation

THE NEWS MINUTE · 29 JUL 2024

Tamil Nadu: Show cause notice issued to 295 colleges over faculty irregularities

Anna University issued show cause notices to 295 engineering colleges asking why the same teachers were listed as full-time at several institutions.

How we used it: Shows the follow-up an officer needs: notice, reply, due date. AnuPalan tracks each as an undertaking.

https://www.thenewsminute.com/amp/story/tamil-nadu/tamil-nadu-show-cause-notice-issued-to-295-colleges-over-faculty-irregularities

CAREERS360 · 28 JUN 2025

Ahead of TNEA Counselling 2025, group flags 13,861 'suspicious faculty' at 224 Tamil Nadu engineering colleges

The group flagged about 13,861 faculty records that appeared manipulated, and blamed the affiliation process and AICTE's inspection system.

How we used it: Why AnuPalan compares records across colleges automatically instead of relying on visits.

https://news.careers360.com/tnea-counselling-2025-tamil-nadu-13861-dummy-faculty-224-engineering-college-anna-university-aicte-cai-arappor-iyakkam-mk-stalin

THE COMMUNE · 17 NOV 2025

DVAC Files FIR Against Anna University Officials And Four Colleges In 'Ghost Faculty' Scam

Vigilance registered a case against 10 university officials and four colleges, alleging lists of teachers already employed elsewhere were approved despite irregularities.

How we used it: Why every approval step in AnuPalan is written to a tamper-evident audit trail.

https://thecommunemag.com/dvac-files-fir-against-anna-university-officials-and-four-colleges-in-ghost-faculty-scam/

CAREERS360 · 29 OCT 2019

Will penalise institutions sharing faculty: AICTE Chairman

The AICTE chairman said many institutions were caught listing the same faculty at two places and warned they would be penalised or closed.

How we used it: Shows the issue is long-running and that AICTE already treats it as serious.

https://news.careers360.com/will-penalise-institutions-sharing-faculty-aicte-chairman

04 Approvals lapsing and institutions closing

Why AnuPalan tracks renewals and deadlines with T-90 and T-30 alerts.

CAREERS360 · 3 AUG 2026

135 AICTE-approved technical institutions closed in 2026-27; 342 shut in three years: Govt

Closures rose from 83 (2024-25) to 124 (2025-26) and 135 (2026-27), 342 in three years, while enrolment grew from 22.1 lakh to 29.6 lakh.

How we used it: Scale: more institutions changing status every year means more cases to track.

https://news.careers360.com/342-aicte-approved-technical-institutions-shut-lok-sabha-reply/amp

CAREERS360 · 5 JUL 2026

AICTE: 58 engineering and technical colleges shut in academic year 2025-26

58 engineering and technical colleges closed in 2025-26, with over 950 courses shut; Uttar Pradesh and Maharashtra had 12 each.

How we used it: Supports status tracking per course, not just per college.

https://news.careers360.com/aicte-58-engineering-technical-colleges-closed-academic-year-2025-26-btech-admission

CAREERS360 · 6 MAR 2022

AICTE closes 97 institutes for not seeking extension of approval for past 6 years

AICTE treated 97 institutes as closed because they had not applied for extension of approval in six years.

How we used it: Example on slide 2: a lapse that ran for years. AnuPalan alerts at 90 and 30 days.

https://news.careers360.com/aicte-closes-97-institutes-for-not-seeking-extension-of-approval-for-past-6-year-objection-within-15-days-aicte-india-org

05 Rules and official sources

The rules AnuPalan's workflow follows.

AICTE

AICTE Approval Process Handbook 2024-27

Why it matters: The stages AnuPalan routes through: scrutiny, Expert Visit Committee, SAC/SHC, timelines and documents.

https://aicte.gov.in/approal-process-2024-2027

AICTE

AICTE Approval Process 2025-26

Why it matters: The current year's notices and handbook.

https://www.aicte-india.org/bureaus/approval/approval-process-2025-26

HOGAN LOVELLS · 17 NOV 2025

India's Digital Personal Data Protection Act 2023 brought into force

Why it matters: DPDP Rules notified 13 Nov 2025; core obligations (notice, consent, security, breach intimation) apply from 13 May 2027. AnuPalan hides complainants and encrypts data.

https://www.hlc.com/en/publications/indias-digital-personal-data-protection-act-2023-brought-into-force-

AICTE / SMART INDIA HACKATHON 2026

SIH26262 problem statement

Why it matters: Requirements A to F: workflow routing, dashboards, EVC scheduling, undertakings, document validity alerts, notifications and escalation.

GOVERNMENT OF INDIA

AICTE Act, 1987 · National Building Code 2016 · National Education Policy 2020

Why it matters: Legal basis for AICTE's role, structural safety norms, and the push for quality and transparency.

06 Technology

Tools in the working prototype and the production plan.

PEOPLE MATTERS · 6 JUL 2023

PM Modi unveils India's homegrown AI platform, Bhashini

Role in AnuPalan: Bhashini offers 1,000+ pre-trained models and real-time speech translation. Planned for AnuPalan's voice complaints in Indian languages.

https://www.peoplematters.in/news/technology/pm-modi-unveils-indias-homegrown-ai-platform-bhashini-heres-all-you-need-to-know-38360

CONVERGENCE INDIA · 6 JAN 2025

Bhashini reaches 100 million monthly inferences with over 7 lakh downloads

Role in AnuPalan: 17 Indian languages, 300+ models: proof it works at national scale.

https://convergenceindia.org/industry-news/technology/bhashini-indias-language-translation-app-reaches-100-million-monthly-inferences-with-over-7-lakh-downloads-118906

CAMUNDA DOCS

Camunda 8: BPMN and DMN workflow engine

Role in AnuPalan: Production engine for approval routing and timers.

https://docs.camunda.io/

GITHUB

Tesseract OCR

Role in AnuPalan: Reads names and validity dates from uploaded certificates.

https://github.com/tesseract-ocr/tesseract

GOOGLE / HUGGING FACE

MuRIL: multilingual model for Indian languages

Role in AnuPalan: Planned threat-level classifier for complaints in Hindi and other languages.

https://huggingface.co/google/muril-base-cased

GITHUB

YOLOv8 (Ultralytics)

Role in AnuPalan: Planned crack detection in inspection photos.

https://github.com/ultralytics/ultralytics

FASTAPI DOCS

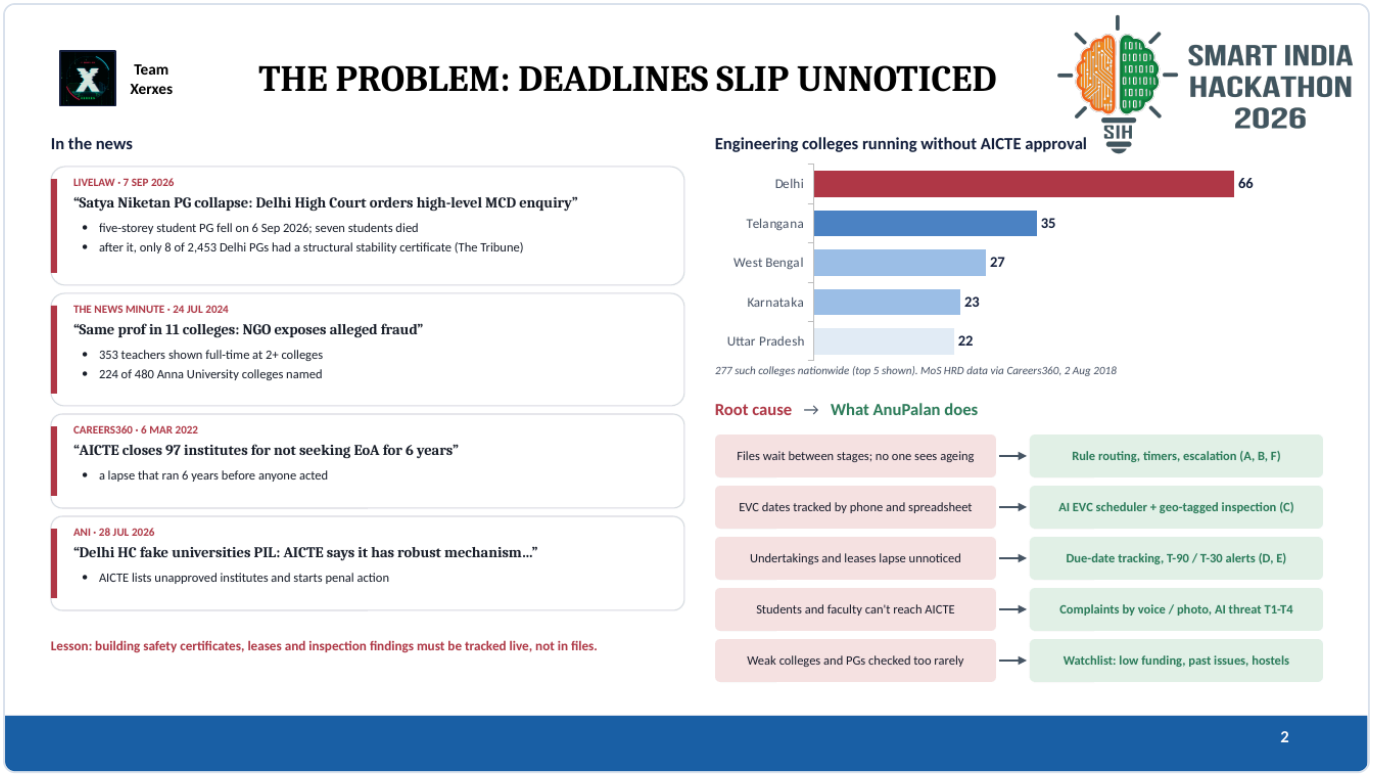
FastAPI

Role in AnuPalan: The working prototype's server.

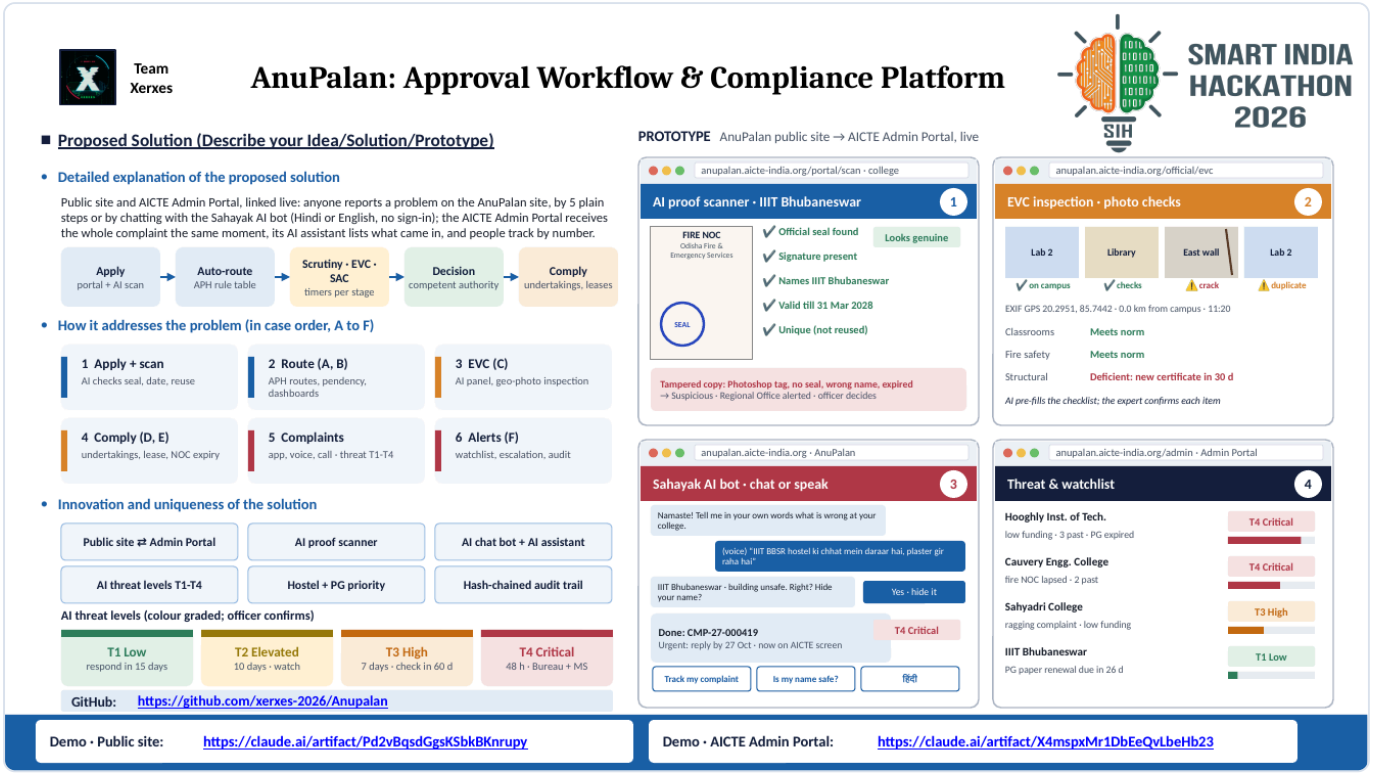
https://fastapi.tiangolo.com/

07 Project diagrams

The diagrams and examples from our deck and blueprint.



Problem slide: news examples, unapproved-college chart and root causes

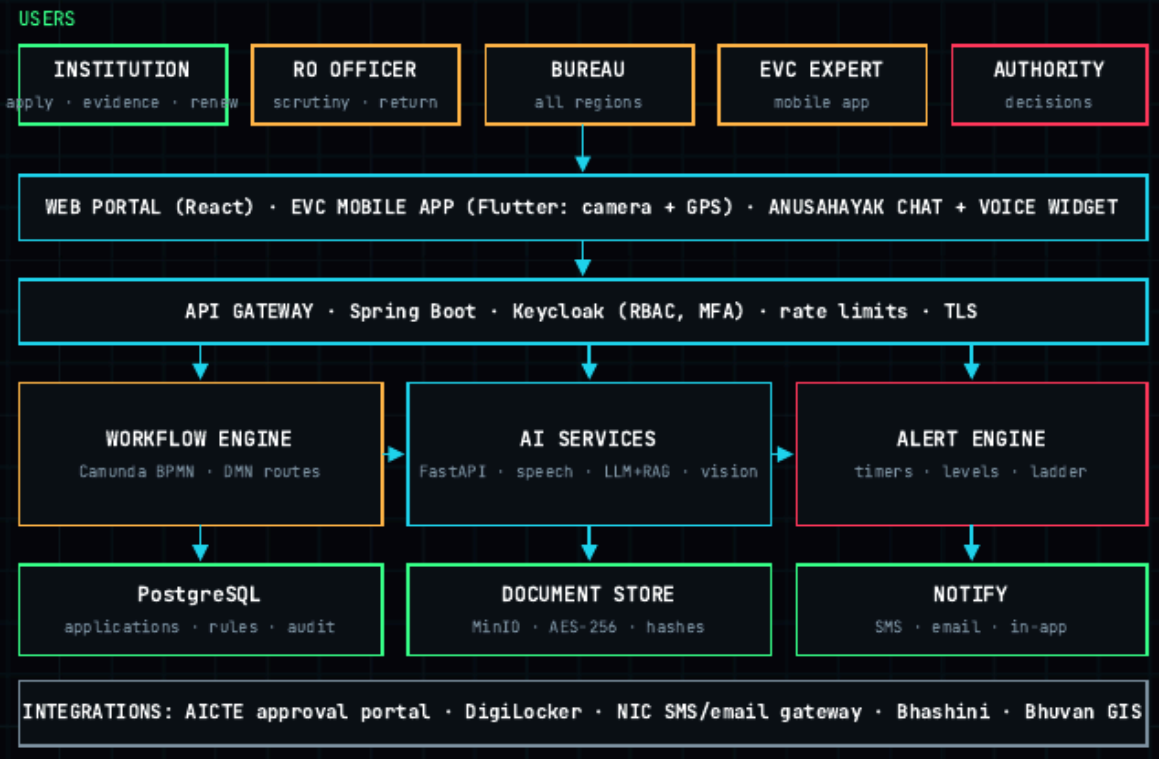


Prototype slide: AI proof scanner, EVC photo checks, Sahayak bot, watchlist

// 02

SYSTEM MAP

Five user roles, one gateway, three engines. The workflow engine owns state; the alert engine watches every timer; AI services only suggest.



Workflow engine

Holds each application's route and current stage. Completing a stage fires the next one. Returns need a reason code.

learn: Camunda 8, BPMN, DMN

Alert engine

Runs every night and on every change. Compares days left with thresholds, sets the colour level, notifies, escalates.

learn: cron, Quartz, Kafka

AI services

Assistant, EVC scheduler, photo checks, OCR, delay-risk model. Stateless; every output is shown to an officer.

learn: FastAPI, Whisper, YOLO

System map

// 03

PUBLIC SITE + ADMIN PORTAL

People put information in on AnuPalan; AICTE acts on it in the Admin Portal. Same API and database: a new complaint pops up on the Admin Portal in full within seconds.



	AICTE ADMIN PORTAL (AICTE ACTS)	ANUPALAN PUBLIC SITE (INPUT)
Address	/official/ · navy header	/ complaint site (no sign-in) · /portal/ college + campus sign-in
Who	Regional Officers, Approval + Monitoring Bureau, Competent Authority, Admin, EVC experts	Anyone: students, parents, teachers, staff, public (complaint site) · colleges such as IIIT Bhubaneswar (sign-in)
AI help	AnuSahayak: "new complaints today" → list, critical first	Sahayak bot: report by chat or voice, track any number
Report flow	1 pick the college · 2 pick the problem (9 picture tiles) · 3 write or speak, add a photo · 4 who you are, name hidden, optional mobile · 5 check and send → complaint number + reply date in plain words. Only officers see the mobile number.	
Main work	route, scrutinise, schedule EVC, inspect, verify, escalate, decide, report, audit	apply, scan + submit proofs, upload evidence, renew documents, complain by app / voice / call, track
Live link	Each site checks `/api/pulse` every 6 s; a new complaint pops up on the Admin Portal in full; the officer's action shows on the AnuPalan site and in the bot.	
Track	Anyone tracks a CMP or APP number without sign-in (`/api/track`): steps and AICTE's action, never who complained.	
Guard	Wrong-site sign-in is refused with a link to the right one (403 + hint).	

ROLE HOME DASHBOARD

- Greeting, date and a one-line status: "Red and orange first".
- KPI tiles, each with a colour stripe and a word: Good / Watch / Act now.
- "Needs your attention": tasks sorted red → green, then by due date, each with an Open button.
- Quick actions for that role, recent activity, AnuSahayak one tap away.

ONE COLOUR SCALE EVERYWHERE

L1 GREEN — on track · valid · T1
L2 YELLOW — due soon · T2
L3 ORANGE — overdue · T3
L4 RED — escalated · expired · T4

Same colours on tiles, menu badges, tables, chips and the legend under the menu. Blue is kept for buttons and information only. Light and dark themes.

MADE TO BE USED AFTER THE HACKATHON

Grouped menu (Overview · Case work · Compliance · Campus voice · Records), global search with the "/" key, plain-language errors that say what to do, password change, phone layout, `run.sh` / `run.bat` start, `.env` settings and API docs at `/docs` for other AICTE systems.

Public site and AICTE Admin Portal, linked live

// 04

CASE JOURNEY (IN ORDER)

The rest of this blueprint follows this order. Each step names who acts, what the AI does, and which alert guards it.

#	STEP	WHO	AI / AUTOMATION	GUARD
1	Apply	Institution	portal form, document list from APH	—
2	AI proof scan	Institution → RO	seal, signature, name, validity date, reuse hash, editing tags	suspicious → RO alert
3	Auto-route + scrutiny	RO officer	DMN route table, stage timer, case summary	T-5 d reminder, overdue → RD
4	EVC scheduling	Bureau	AI ranks experts, picks date within 21 / 30 d	T-2 reminder, delay escalation
5	Inspection	EVC expert	EXIF GPS + time, duplicate, blur, crack checks; checklist pre-fill	crack → undertaking
6	SAC/SHC + decision	Competent authority	none: humans decide	decision timer
7	Undertakings	Institution → RO	evidence scan, verify / reject	T-15 reminder, escalation
8	Lease + documents	Institution → RO	OCR reads new expiry	T-90 / T-30 / expired
9	Complaints	Students, staff, public	app, voice or phone call; category + threat T1-T4	15 / 10 / 7 d / 48 h
10	Threat + watchlist	Bureau	risk score 0-100, periodic checks	30 / 60 / 90 / 365 d
11	Hostels + PGs	RO	certificate expiry, old-building flag	expired → T4
12	Alerts + audit	Everyone	escalation ladder, reports, hash chain	verify chain

WORKED EXAMPLE: IIIT BHUBANESWAR (ILLUSTRATIVE DEMO RECORDS)

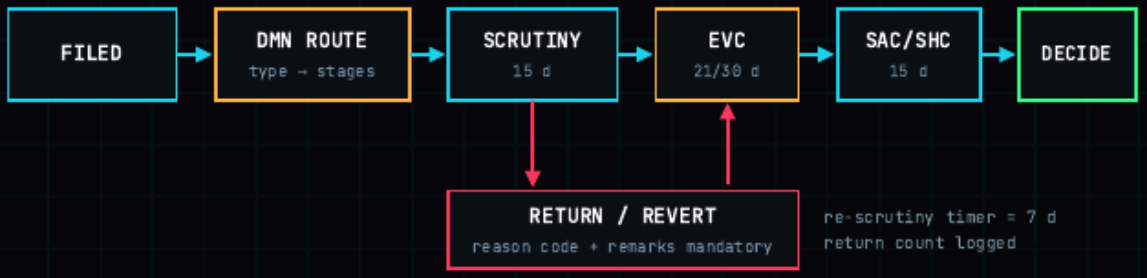
- **Day 0** · APP-27-1160, Increase in Intake (B.Tech CSE), Kolkata RO. Fire NOC scanned: `LOOKS GENUINE` .
- **Day 1-7** · Routed "Scrutiny + EVC"; scrutiny done; case moves to the EVC queue automatically.
- **Day 8** · AI panel: Dr. S. Mohanty (NIT Rourkela) + Prof. A. Das (Jadavpur); visit inside 30 days.
- **Visit** · photos at 20.295 N, 85.744 E, on campus; checklist confirmed; decision by the authority.
- **After** · UT-0330 faculty orders due in 20 d; DOC-5508 fire NOC T-45 reminder; PG near Patia renewal alert.
- **Any time** · a student calls the helpline about the hostel; complaint gets a number and a threat level.

Case journey from application to decision

// 06

WORKFLOW ROUTING (A)

The route is picked by a decision table, not by a person. Change the table, and new applications follow the new route the same day.



ROUTE TABLE (EDITABLE IN RULES & TIMELINES)

APPLICATION TYPE	ROUTE	STAGES
New Institution	Scrutiny + EVC + SAC/SHC	Scrutiny > EVC > SAC/SHC > Decision
EoA (no change)	Only Scrutiny	Scrutiny > Decision
Increase in Intake	Scrutiny + EVC	Scrutiny > EVC > Decision
Change of Site	EVC + SAC/SHC	EVC > SAC/SHC > Decision
Restoration of Intake	Only EVC	EVC > Decision
New Programme	Scrutiny + EVC	Scrutiny > EVC > Decision

AUTO-ADVANCE

- Officer marks the stage complete; the engine moves the case to the next stage and restarts the timer.
- EVC cannot complete without a report with at least 2 valid geo-tagged photos and 4 checklist items confirmed.
- The last stage records the authority's decision and closes the case.

RETURN / REVERT

- Two choices: return to institution (deficiency) or revert to previous stage.
- Reason codes: deficiency, clarification, further verification. Remarks are mandatory.
- Institution is notified by email + SMS with the reason.

ROUTINE AUTOMATION

AI case summary (stage, days left, open undertakings, expiring documents, EVC status) on every file. Standard letters drafted from templates; the officer reviews and sends. Both are logged.

Workflow routing (requirements A and B)

// 14

THREAT LEVELS T1–T4

One colour scale for everything AnuSahayak watches: complaints, institutions, hostels and PGs. The AI proposes the level and shows its reasons; the officer confirms.

LEVEL	MEANING	RESPONSE	WHO IS ALERTED
T1 LOW	routine issue, no safety risk	15 days	RO help desk
T2 ELEVATED	service failure or a repeat	10 days	RO officer; institution added to watch
T3 HIGH	safety, harassment or legal breach	7 days	RO + Monitoring Bureau; check within 60 d
T4 CRITICAL	danger to life, or unapproved institution	48 hours	Bureau Head + Member Secretary

HOW A COMPLAINT'S LEVEL IS SET

```
level = base(category) +1 if hostel / PG resident +1 if ≥2 similar in 30 d +1 if institution has ≥2 past violations (max T4)
```

Example: "Hamare PG ki deewar mein daraar" → structural (T3) + PG (+1) = T4 . Response in 48 h, Bureau and Member Secretary alerted.

INSTITUTION RISK SCORE (0–100)

SIGNAL	POINTS
open complaints	T1 +2 · T2 +5 · T3 +10 · T4 +20 each
low funding	+10 if below ₹40,000 per student per year
past violations	+5 each, up to +15
hostel / PG safety paper	+15 expired · +6 expiring within 30 d
old PG building	+4 if older than 40 years
overdue undertakings	+5 each
expired documents	+8 each

SCORE → COLOUR

T1 <15 | T2 15–29 | T3 30–49 | T4 50+

Same colours on dashboards, chips, map pins, emails and SMS headers.

EXPLAINABLE

Every level shows its reasons, e.g. "low funding · 3 past issues · PG paper expired". Production: MuRIL text classifier + gradient boosting; prototype: keyword rules and the formula above.

Threat levels T1 to T4

// 17

STACK DOSSIER

Every tool, why we picked it, and what to read to explain it to a judge.

React + Vite
Web portal and dashboards; role-based routes.
`learn: react.dev, TanStack Query`

Flutter
EVC mobile app: camera, GPS, offline queue, upload on signal.
`learn: camera, geolocator plugins`

Spring Boot
Core API: applications, undertakings, documents, reports.
`learn: Spring Data JPA, validation`

Keycloak
Login, MFA, roles (RO, Bureau, EVC, Institution, Authority, Admin).
`learn: OIDC, realm roles`

Camunda 8 (BPMN + DMN)
Routes as decision tables; stages as BPMN tasks with timer events.
`learn: DMN hit policies, boundary timers`

PostgreSQL + pgvector
All records, audit chain, APH embeddings for RAG.
`learn: JSONB, triggers, pgvector`

MinIO
Encrypted document and photo store; SHA-256 stored per file.
`learn: S3 API, server-side encryption`

FastAPI
AI microservices: assistant, scheduler, photo checks, OCR, risk.
`learn: pydantic, async endpoints`

Bhashini + Whisper
Speech-to-text and text-to-speech in Indian languages.
`learn: bhashini.gov.in ULCA APIs`

Tesseract + dateparser
OCR on renewals; reads expiry dates.
`learn: pytesseract, regex`

Ultralytics YOLOv8
Scene tagging and crack segmentation on inspection photos.
`learn: public crack datasets, fine-tuning`

scikit-learn
Delay-risk model: stage, age, officer load, returns → P(overdue).
`learn: gradient boosting, SHAP`

Kafka / Quartz
Event bus and nightly alert sweep.
`learn: topics, cron triggers`

NIC SMS + email
Official notification channels; DLT-registered templates.
`learn: DLT templates, SMTP`

Technology stack

08 Live demo and code

Demo · Public site: https://claude.ai/artifact/Pd2vBqsdGgsKSbkBKnrupy

Demo · AICTE Admin Portal: https://claude.ai/artifact/X4mspxMr1DbEeQvLbeHb23

GitHub: https://github.com/xerxes-2026/Anupalan

Summaries are our own, written from each article; all articles belong to their publishers. AnuPalan is a student prototype for SIH 2026 and not an official AICTE system. For research purposes only.